Natural Language Processing (NLP)

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1. What is Natural Language Processing?

Natural Language Processing (NLP) is the field of AI that enables computers to understand, interpret, and generate human language — both spoken and written. When we listen to words and form comprehension, we are doing NLP naturally. When we ask a computer to do the same, that is NLP.

Unstructured Text

How humans naturally speak and write. Informal, contextual, and ambiguous to a computer.

Example: 'Add eggs and milk to my shopping list.'

You and I understand this perfectly, but a computer cannot directly act on it.

Structured Data

A formal, organised representation that a computer can process and act upon.

Example: A shopping list object with child elements: item='eggs', item='milk'.

Machines can read, store, and query structured data reliably.

The job of NLP is to translate between unstructured and structured data. NLP sits in the middle as the bridge between human language and machine-readable information.

2. NLU vs NLG — Two Directions of NLP

NLU — Natural Language Understanding

Direction: Unstructured → Structured

The computer reads or hears human language and converts it into a structured, machine-processable form.

Example: Parsing 'Add eggs and milk to my list' into a structured shopping list object.

This is the primary focus of most NLP pipelines.

NLG — Natural Language Generation

Direction: Structured → Unstructured

The computer takes structured data and converts it into natural human language as output.

Example: Converting a weather data object into the sentence 'It will be sunny and 28 degrees tomorrow.'

Used in report generation, chatbot responses, and summarisation.

3. Key Use Cases of NLP

Machine Translation	Translating text between languages requires understanding the full context and sentence structure — not just word-for-word substitution. Classic failure: translating 'the spirit is willing but the flesh is weak' (English → Russian → English) returns 'the vodka is good but the meat is rotten'.
Virtual Assistants & Chatbots	Virtual assistants (e.g. Siri, Alexa) take spoken human utterances and derive commands to execute. Chatbots use written language to traverse a decision tree and take appropriate actions. Both rely heavily on NLU to interpret user intent.
Sentiment Analysis	Analyse text (emails, product reviews, social media) to determine the sentiment expressed. Identifies whether text is positive or negative, genuine or sarcastic. Widely used in customer experience, brand monitoring, and market research.
Spam Detection	Examine email content to classify it as genuine or spam. Indicators include: overused words, poor grammar, inappropriate urgency claims, suspicious links. NLP extracts these signals to make classification decisions.

4. The NLP Processing Pipeline

NLP is not a single algorithm — it is a **bag of tools** that can be applied in sequence to transform unstructured text into structured, meaningful data. Each stage enriches the tokens with more information:

Input **Unstructured Text**	Raw human speech or written text. Example: 'Add eggs and milk to my shopping list.' Spoken audio is first converted to text via a speech-to-text algorithm before NLP processing begins.
Step 1 **Tokenization**	Break the input string into individual chunks called tokens. Each word (or meaningful unit) becomes one token. Example: 'Add | eggs | and | milk | to | my | shopping | list' → 8 tokens. All subsequent stages operate one token at a time.
Step 2 **Stemming**	Derive the base word stem for each token by removing prefixes and suffixes and normalising tense. Example: 'running', 'runs', 'ran' → all stem to 'run'. Limitation: does not work well for all words (e.g. 'university' and 'universal' do not correctly stem to 'universe').
Step 3 **Lemmatization**	Find the root (lemma) of a token using dictionary definitions and morphological analysis. More accurate than stemming for irregular words. Example: 'better' → lemma is 'good' (not 'bet' as stemming would give). Use lemmatization when accuracy of root form matters.
Step 4 **Part of Speech Tagging**	Determine the grammatical role of each token in the context of the sentence. The same word can be different parts of speech depending on usage. Example: 'make' in 'I will make dinner' → verb. 'make' in 'What make is your laptop?' → noun.

Step 5 **Named Entity Recognition (NER)**	Identify whether a token refers to a real-world entity and classify it. Example: 'Arizona' → entity type: US State. 'Ralph' → entity type: Person's name. Helps machines understand references to specific people, places, organisations, dates, etc.
Output **Structured Data**	The processed tokens, now enriched with grammatical and semantic information, are assembled into a structured representation that a computer can act upon. Example: a shopping list XML with item elements for 'eggs' and 'milk'.

5. Stemming vs Lemmatization — Comparison

Both techniques aim to find the root form of a word, but they differ in approach and accuracy:

Token	Stemming Result	Lemmatization Result	Note
running / runs / ran	run ✓	run ✓	Both work
university / universal	univers ✗	university / universal ✓	Lemma preferred
better	bet ✗	good ✓	Lemma preferred

Rule of thumb: Use stemming for speed when approximate root forms are acceptable. Use lemmatization when accuracy of meaning matters — it is slower but linguistically correct.

6. Part of Speech Tagging & Named Entity Recognition

Part of Speech (POS) Tagging

POS tagging determines the grammatical role of each token within the context of its sentence. The same word can function as different parts of speech depending on how it is used:

Sentence	Token	POS Tag
'I am going to make dinner'	**make**	Verb
'What make is your laptop?'	**make**	Noun

Named Entity Recognition (NER)

NER identifies whether a token refers to a real-world named entity and classifies its type. This helps machines understand references to specific people, places, organisations, and more:

Token	Entity Type	Example Use
Arizona	US State / Location	Geography, address parsing
Ralph	Person's Name	Contact resolution, HR systems

Monday	Date / Time	Calendar, scheduling apps

7. Key Takeaways

1	NLP enables computers to understand and generate human language, bridging unstructured text and structured data.
2	NLU converts human language into structured data (unstructured → structured). NLG does the reverse.
3	NLP is a bag of tools applied in a pipeline — not a single algorithm.
4	Tokenization is the first step: break input text into individual tokens for further processing.
5	Stemming finds a word's base form by removing affixes; lemmatization uses dictionary lookup for the true root — more accurate.
6	Part of Speech tagging identifies the grammatical role of each token based on its context in the sentence.
7	Named Entity Recognition (NER) identifies real-world entities (people, places, dates) associated with tokens.
8	NLP powers many everyday applications: machine translation, virtual assistants, sentiment analysis, and spam detection.